WORKED SOLUTION

STEP III 2024, Question 3

Pure • Monotonic Functions and Derivatives

STEP III 2024, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything hangs on one derivative. When you differentiate g and put it over a common denominator, the numerator collapses to a *linear* expression in x, so its sign is easy to read off. The link to part (iii) is that g is exactly the derivative of $\ln f$. So the sign of g, which the sketches in (ii) make visible, decides whether f rises or falls.

Solution

Part (i)

Since $x > 0$, write $\ln(1+\frac{1}{x}) = \ln(x+1) - \ln x$. Then

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln\left(1+\frac{1}{x}\right) = \frac{1}{x+1} - \frac{1}{x} = -\frac{1}{x(x+1)} = -\frac{x(x+1)}{x^2(x+1)^2},$$

and, by the quotient rule with denominator x^2+x,

$$\frac{\mathrm{d}}{\mathrm{d}x}\,\frac{x+c}{x^2+x} = \frac{(x^2+x)-(x+c)(2x+1)}{x^2(x+1)^2} = \frac{-x^2-2cx-c}{x^2(x+1)^2}.$$

Subtracting, the x^2 terms cancel:

$$\mathrm{g}'(x) = \frac{-x^2-x+x^2+2cx+c}{x^2(x+1)^2} = \frac{(2c-1)x+c}{x^2(x+1)^2}.$$

The denominator is positive, so the sign of $\mathrm{g}'(x)$ is the sign of $N(x) = (2c-1)x + c$.

(a) If $c \geq \frac{1}{2}$ then $(2c-1)x \geq 0$ and $c \geq \frac{1}{2} > 0$, so $N(x) > 0$ for every $x > 0$. Hence the gradient is positive for all $x > 0$.

(b) If $0 \leq c < \frac{1}{2}$ then $1 - 2c > 0$, and

$$N(x) < 0 \iff (1-2c)x > c \iff x > \frac{c}{1-2c}.$$

So the gradient is negative exactly when $x > \dfrac{c}{1-2c}$. (When $c = 0$ this is every $x > 0$.) Write $x_0 = \dfrac{c}{1-2c}$. For $c > 0$, g is increasing on $(0, x_0]$ and decreasing on $[x_0, \infty)$.

Part (ii)

We also need the behaviour for large x. Both terms of g tend to 0, so $g(x) \to 0$, but the sketch should show which side it approaches from. Using $\ln(1+t) = t - \frac{1}{2}t^2 + \cdots$ with $t = \frac{1}{x}$, and $\dfrac{x+c}{x(x+1)} = \left(\dfrac{1}{x} + \dfrac{c}{x^2}\right)\left(1 - \dfrac{1}{x} + \cdots\right) = \dfrac{1}{x} + \dfrac{c-1}{x^2} + \cdots$, we get

$$g(x) = \frac{\frac{1}{2} - c}{x^2} + \cdots \quad \text{for large } x.$$

Ask yourself If g is increasing and tends to 0, which side of the axis must it be on?

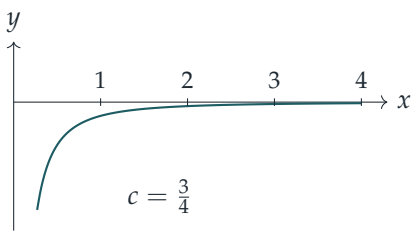
It must be below. If g is strictly increasing then, for any x, $g(x) < g(y)$ for all $y > x$, and letting $y \to \infty$ gives $g(x) < 0$ (strict, because $g(x) < g(x+1) \leq 0$). In the same way, a strictly decreasing function that tends to 0 is positive.

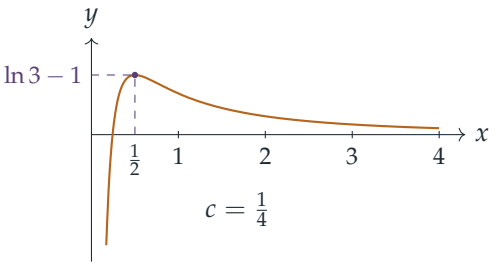
(a) $c = \frac{3}{4}$**.** By (i)(a), g is increasing throughout. It rises from $-\infty$ (as $x \to 0$) towards 0 and stays below the axis. It approaches like $-\frac{1}{4x^2}$, so it never crosses and has no turning point. For example, $g(1) = \ln 2 - \frac{7}{8} \approx -0.18$.

(b) $c = \frac{1}{4}$**.** Here $x_0 = \dfrac{1/4}{1/2} = \frac{1}{2}$. The curve rises from $-\infty$ to a maximum at $x = \frac{1}{2}$, where

$$g(\tfrac{1}{2}) = \ln 3 - \frac{3/4}{3/4} = \ln 3 - 1 \approx 0.099 > 0.$$

It then decreases towards 0 from above, like $\frac{1}{4x^2}$. On $[\frac{1}{2}, \infty)$ it is decreasing to 0, so it is positive there. On $(0, \frac{1}{2})$ it is increasing from $-\infty$ to a positive value, so it crosses the axis exactly once, at a point a little below $\frac{1}{4}$ (numerically $x \approx 0.243$).





(The vertical scales differ: the right-hand graph is stretched to make the small positive hump visible. In both graphs the y-axis is an asymptote.)

Part (iii)

Since $f(x) > 0$ for $x > 0$, we can take logarithms:

$$\ln f(x) = (x + c)\ln\left(1 + \frac{1}{x}\right).$$

Differentiating with the product rule, and using the derivative from (i),

$$\frac{f'(x)}{f(x)} = \ln\left(1 + \frac{1}{x}\right) + (x + c)\cdot\left(-\frac{1}{x(x+1)}\right) = g(x).$$

So $f'(x) = f(x)\,g(x)$, and because $f(x) > 0$, f' has the same sign as g. Note also that $g(x) \to 0$ as $x \to \infty$ for every c, since both of its terms tend to 0.

(a) $c \geq \frac{1}{2}$**.** By (i)(a), g is strictly increasing on $x > 0$ and tends to 0. By the argument after the box in (ii), $g(x) < 0$ for all $x > 0$. Hence $f'(x) < 0$, and f is decreasing.

(b) $0 < c < \frac{1}{2}$**.** By (i)(b), g is strictly decreasing on $[x_0, \infty)$ and tends to 0, so $g > 0$ there; in particular $g(x_0) > 0$. On $(0, x_0]$, g is continuous and strictly increasing, with $g(x) \to -\infty$ as $x \to 0$ (given, as $c > 0$). It therefore takes the value 0 at exactly one point $x_1 \in (0, x_0)$, with $g < 0$ on $(0, x_1)$ and $g > 0$ on (x_1, ∞). So $f'(x_1) = 0$, and f' changes sign from negative to positive there. Hence f has a turning point (a minimum) at x_1.

> **?** **Ask yourself** Why isn't $f'(x_1) = 0$ on its own enough?

A stationary point need not be a turning point (think of $y = x^3$ at the origin). What makes x_1 a turning point is the change of sign of f', and that is exactly what the shape of g supplies.

(c) $c = 0$**.** Now $g'(x) = \dfrac{-x}{x^2(x+1)^2} < 0$ for all $x > 0$, so g is strictly decreasing and tends to 0. Hence $g(x) > 0$ for all $x > 0$, so $f'(x) > 0$, and $f(x) = \left(1 + \frac{1}{x}\right)^x$ is increasing.

Checks

- Differentiate g numerically at a point, say $x = 1$, $c = \frac{1}{4}$: the formula gives $g'(1) = \frac{-1/2+1/4}{4} = -\frac{1}{16}$, which matches a difference quotient.
- $c = \frac{1}{4}$: $g(0.2) \approx -0.083$ and $g(0.3) \approx 0.056$, consistent with the single crossing near 0.243. For $c = \frac{3}{4}$, $g(3) \approx -0.025$, still just below the axis.
- In (iii)(b) with $c = \frac{1}{4}$, the minimum of f is at the same $x \approx 0.243$, and a computer check of f on a fine grid finds exactly one turning point for several values of c in $(0, \frac{1}{2})$.
- For $c = 0$, $f(x) = (1 + \frac{1}{x})^x$ increases to e; for $c = 1$, $(1 + \frac{1}{x})^{x+1}$ decreases to e. That's the familiar pair of sequences that squeeze e.

Common slips

- Concluding $g < 0$ from "g increasing" alone. You also need the limit $g \to 0$ as $x \to \infty$.
- In (ii)(b), drawing the curve as always negative. The maximum $\ln 3 - 1$ is small but positive, so the curve must cross the axis and come back down to it from above.
- Differentiating $(1 + \frac{1}{x})^{x+c}$ as if the exponent were constant. Take logarithms first.

Beyond STEP

The fact that $(1 + \frac{1}{x})^x$ increases and $(1 + \frac{1}{x})^{x+1}$ decreases, both towards e, is a classic first-year way of proving that the limit defining e exists. It is an instance of the monotone convergence theorem. This question finds the precise threshold $c = \frac{1}{2}$ where the behaviour switches, and the expansion $g(x) \approx (\frac{1}{2} - c)/x^2$ explains why. For more on monotone sequences and limits, try Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.